

Estimation of Residual Intersectional Effects

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Background

In the MAIHDA framework, the residual intersectional effect represents the extent to which each stratum deviates from the expected outcome under the fixed-effects model.

The residual intersectional effect is captured by u_j on the log-odds scale, or by calculating the difference in predicted probabilities: total probability (random + fixed) minus probability (fixed) in percentage points.

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*Graphs attached.

Log-odds Scale Estimation

First, stratum-specific random effects and their conditional standard errors were extracted directly from the fitted multilevel model. This approach operates on the model's native (logit) scale. Statistical significance was defined as intervals not overlapping zero. This representation is statistically closest to the underlying model specification, therefore the primary inferential scale. However, given that this approach is based on simulation, the results vary consistently depending on number of simulations performed. The default of `REsim()` is $n = 200$.

1. Using `REsim()` (logit scale).
 - Base R function that uses simulation to get:
 - mean random effect
 - sd
 - Approximate 95% CI as mean $\pm 1.96 \times \text{sd}$
1. Using `ranef()` with conditional variance (logit scale).
 - Extract conditional posterior variance directly from `lme4`
 - $\text{SE} = \sqrt{\text{postVar}}$
 - Construct 95% CI as: $u_j \pm 1.96 \times \text{SE}(u_j)$
 - This approach uses model-based conditional variance.

Probability Scale Estimation

Because probability differences are more interpretable, residual effects were also expressed on the probability scale. CI on the probability scale were obtained using two approaches:

1. Simulation-based approximation using 1,000 Monte Carlo draws from the estimated distribution of the random effects *Evan's Tutorial.
 - Duplicate strata 1000 times
 - Add Normal noise around the linear predictor using its SE
 - Transform to the probability scale
 - Compute probability difference
 - This approach creates uncertainty in the linear predictor via Monte Carlos simulation. The CI is calculated as mean +/- 1.96*sd of simulated differences.
2. Analytical transformation of logit-scale intervals.
 - Explicitly build X matrix of the variables in stratum
 - Compute fixed-effects
 - Add the random effect from ranef()
 - Transform to probability
 - Compute difference
 - CIs calculated using total effects mean because treating fixed effects as known.
3. Hybrid.
 1. Extract stratum random effects using ranef()
 2. Compute fixed-effects only
 3. Add the stratum effect to fixed-effects
 4. Convert to probabilities
 5. Take the difference
 6. Compute 95% CI using REsim (simulated distribution)
 7. Construct CIs on the logit scale then convert to probability
 8. Compute difference removing the fixed-effects in probability scale

Comparison of Approaches

Across approaches, point estimates of the residual intersectional effects were highly consistent in magnitude and direction, however some minor differences were observed in:

- The ordering of strata when ranked by residual effect.
- The classification of a small number of borderline strata as significantly significant.

I believe these differences arise from three methodological features:

1. Nonlinearity of the logit transformation:

On the logit scale, significance is assessed relative to $u_j = 0$. On the probability scale, significance is assessed by: $\text{plogis}(\text{fixed-effects} + u_j) - \text{plogis}(\text{fixed-effects}) = 0$. Because of the nonlinear transformation, these are not perfectly equivalence tests, so disagreements are mathematically expected.

2. Monte Carlo variability:

When simulate 1000 times, the CI limits and even mean estimates have sampling variability. If some strata are borderline (CI touching 0), small simulation differences can flip significance. The probability-scale CIs are based on:

3. Small residual variance:

The proportional change in variance ~ 90% indicates that most between-stratum variance is explained by additive effects. The remaining random effects ~ 10% are small, making significance classifications sensitive to small computational differences.

Overall

The graphs both support the conclusion that the differences between strata are largely explained by fixed-effects, with limited evidence of residual intersectional mechanisms.

We had previously thought of performing bootstrapping, to overcome this change of scale of the CIs, however, this would potentially introduce uncertainty in the fixed-effects actually making the CIs wider. Since our question is conditional on the fixed-effect parameters, we can treat these as true and not consider additional uncertainty.

Next Steps

Given the high proportion of variance explained by additive effects and the small magnitude of the remaining residual variance, I think our interpretation should focus on:

- The magnitude and distribution of residual effects,
- The overall pattern of deviation from fixed-effects,
- The absence of large intersectional departures.

Classifying strata as significant or non-significant in this case could be erroneous as the statistical significance is sensitive to the transformation of scale and simulation procedure, and because most intervals are borderline around zero.